

AlphaForge

A quantitative alpha-research platform: synthetic-to-real data, cross-sectional factor engine, evolutionary multi-agent RL, and a deflation-aware rigor layer.

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Components: JS frontend · Python alpha engine · MARL framework · Live paper-trading execution

Scale: 3 Python backend services · 618+ tests · JS/Python parity to 10 dp · 10 years of real OHLCV for 50 US large-caps

This document describes the architecture of the AlphaForge platform, the data pipeline that underpins it, the strategies it evaluates, and — critically — what rigorous statistical testing has revealed about those strategies. The goal is to present an honest picture of both what has been built and what the results mean. Deflation-aware reporting (Bailey & López de Prado, 2014), bootstrap confidence intervals, and baseline-excess decomposition are first-class outputs, not afterthoughts.

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1. Executive summary

AlphaForge is a full quantitative research platform built end-to-end: data acquisition and storage, factor construction, cross-sectional backtesting, mean-variance portfolio optimisation, an evolutionary multi-agent reinforcement-learning framework, and live paper-trading execution. The system was designed to answer one question: *can a population of RL agents evolve trading strategies that generalise to unseen market regimes?*

The honest answer, after building the rigor layer: not yet — but the platform that was built to evaluate that question is itself the research artifact. The project now produces two deflation-aware research reports (single-factor and MARL) that apply Bailey & López de Prado's Deflated Sharpe Ratio, stationary-bootstrap Sharpe CIs, and baseline-excess decomposition against an equal-weight benchmark. These reports demonstrate the statistical hygiene a quant researcher is expected to apply to their own work.

Key numbers

- 50 US large-cap universe, 2016–2025 (10 years of point-in-time-validated daily OHLCV).
- 5 cross-sectional factors in the JS-parity set; 6 total.
- 3 Python backend services (alpha engine, MARL, execution), each with its own FastAPI.
- 618+ tests; JS/Python numerical parity verified to 10 decimal places.
- Headline finding: no single factor clears Deflated Sharpe > 0.95 on this universe.
- Headline MARL finding: OOS stability Sharpe 0.72 $\rightarrow$ DSR 0.04; zero of 76 reward-mix trials beat equal-weight.

Negative findings, properly documented, are a deliberately chosen headline. This report explains why the project is stronger, not weaker, for reporting them.

2. Architecture overview

AlphaForge is organised as four loosely-coupled components, each with its own CLAUDE.md, test suite, and (where applicable) FastAPI service:

Component	Role	Tests
JS frontend	Single-page research UI over the alpha engine API.	—
alphaforge-python	Data, factors, backtest, optimiser, FastAPI.	408
alphaforge-marl	Evolutionary multi-agent RL (NSGA-II + MAML + HMM regimes).	122
alphaforge-execution	Live paper-trading loop with Alpaca + yfinance + SQLite.	106

Data flows unidirectionally: **yfinance** $\rightarrow$ **parquet store** $\rightarrow$ **alphaforge-python** (factor scoring, backtest, optimiser) $\rightarrow$ **alphaforge-marl** (imports alpha engine via sys.path, trains agents on real OHLCV windows) $\rightarrow$ **alphaforge-execution** (runs extracted strategies live against Alpaca paper

trading). The parquet store is the single source of truth; only sync_market_data.py and the execution daily loop touch yfinance.

3. Data pipeline (synthetic → real)

The project started on a seeded-PRNG synthetic data layer (Mulberry32 → geometric Brownian motion) for reproducibility and JS/Python parity testing. Over time the primary evaluation substrate migrated to real market data, stored locally as parquet files (one per ticker per year) under data/market/.

Universe manifest

50 US large-caps across Technology, Healthcare, Finance, Consumer, and Energy. Each ticker has a manifested usable_start date that respects IPO / spin-off / restructuring history (e.g. META 2012-05-18, TSLA 2010-06-29, ABBV 2013-01-02). A quarantine subsystem flags parquet files whose validator checks fail and excludes them from training/evaluation.

Why parquet, not a database

Parquet per-ticker-per-year gives $O(1)$ range lookups, zero-install read, and columnar compression. A small MarketDataLoader with an LRU cache makes repeated backtests cheap. Read-only by design; all mutations go through the sync + validator path.

Point-in-time limitations (acknowledged)

The universe is defined as of today, not point-in-time — delisted peers are not included. This biases the long-only equal-weight baseline upward by an estimated 1–2% annualised. Any future capital-allocation-grade result must use a CRSP or Norgate survivorship-bias-free universe.

4. Strategy layer — factors & MARL

4.1 Cross-sectional factors

Six factor implementations, five of which are bit-for-bit identical with the JS frontend (verified to 10 decimal places):

Factor	Formula (JS-parity variant)	Lookback
Momentum (12-1)	$(p[t-21] - p[t-252]) / p[t-252]$	252d
Mean Reversion (5d)	$-(p[t] - p[t-5]) / p[t-5]$	5d
Volume Surge	$(\text{mean}(v[-5:]) - \text{mean}(v[-20:])) / \text{mean}(v[-20:])$	20d
RSI Divergence	$(\text{RSI}[\blacksquare](p) - 50) / 50$	14d
Earnings Drift	$(p[t] - p[t-10]) / p[t-10]$	10d
Low Volatility	$-\text{std}(\text{ret}[-60:])$ (inverse realised vol)	60d

Factors are scored daily per ticker, z-scored cross-sectionally at each rebalance, and fed into a unified long-short simulation engine that reports 9 performance metrics plus OLS attribution.

4.2 MARL framework

The multi-agent RL layer is a population-based trainer with four subsystems:

- **TradingEnv** — Gymnasium env, 57-dim observation, 5 discrete or 10-dim continuous actions, dense reward shaping (rolling Sharpe delta + drawdown penalty + participation incentive).
- **EvolutionaryEngine** — NSGA-II multi-objective selection on (Sharpe, drawdown, turnover), speciation via Jensen-Shannon divergence on probe-state action distributions, adaptive per-parameter mutation.
- **PPOTrainer + MAMLTrainer** — GAE + clipped-surrogate PPO for survivor fine-tuning; FOMAML for fast regime adaptation.
- **RegimeBandit** — HMM regime detector (K-Means init + Baum-Welch EM), Thompson sampling per (regime, agent), capital allocator feeding an EnsemblePolicy.

4.3 Strategy honesty note

All six factors are textbook. They are the *implementation canvas*, not the alpha. The rigor layer (§5) exists precisely to expose the limits of textbook signals on a 50-ticker universe after realistic costs. Next-step strategies (residual momentum, short-interest changes, earnings-call sentiment) are discussed in §10.

5. Research rigor

Two research scripts produce the headline artifacts of the project. Both were built specifically to avoid the two most common failure modes of quant portfolio projects: selection bias and baseline omission.

5.1 Single-factor study ([alphaforge-python/research/factor_study.py](#))

- Spearman IC at horizons {1, 5, 10, 21, 63} days — per-factor, per-horizon
- Quintile-spread backtest (5 buckets, monthly rebalance, 21-day holding)
- Realistic transaction costs: 1 bp commission + 2 bp half-spread + 10 bp × turnover² impact
- Stationary bootstrap (2000 reps, mean block length 21 days) on the net Sharpe
- Deflated Sharpe Ratio across the 5-factor trial set
- Equal-weight long-only and random-long-short (100 seeds) baselines
- Regime split on the best factor by 21-day realised-vol quantiles of the benchmark

5.2 MARL rigor report ([alphaforge-marl/research/marl_rigor.py](#))

- Scans every training.jsonl under ablations, reward-mix sweep, and stability runs
- Enumerates the full generation-level trial count (N = 100 reported; true N is higher)
- Reports the distribution of per-generation val Sharpe across the whole search

- Computes DSR of both the in-sample maximum and the honest OOS stability mean
- Summarises the baseline-excess Sharpe distribution from the reward-mix logs — *the only honest Sharpe*
- Reports seed-to-seed OOS stability

5.3 Logging infrastructure for future runs

compute_performance_metrics in the MARL baselines module now returns daily_returns and nav_series lists alongside scalar metrics; aggregate_metric_dicts concatenates list-valued keys across windows (scalars still averaged). Every future stability / ablation / benchmark run will persist per-day portfolio paths automatically, enabling bootstrap CIs at report time without re-evaluating the environment.

6. Findings — what the numbers actually say

6.1 Single-factor study

Factor	IC t @ h=63	Gross SR	Net SR	DSR
Momentum (12-1)	+7.52	+0.17	+0.11	0.14
Mean Reversion (5d)	-0.29	-0.15	-0.79	0.00
Volume Surge	-0.39	+0.10	-0.86	0.00
RSI Divergence	+2.05	+0.15	-0.53	0.00
Earnings Drift	+0.06	+0.28	-0.37	0.00

Equal-weight long-only baseline: Sharpe +0.92, annual return +17.3%, max DD -37.5%.

Random long-short (100 seeds): mean Sharpe +0.01, 95% CI [-0.47, +0.52]. Any factor whose net Sharpe falls inside this band is statistically indistinguishable from randomness.

6.2 MARL rigor report

Metric	Value
Total trials enumerated	100
Val-Sharpe distribution (max)	+6.74
Val-Sharpe distribution (mean)	+0.17
In-sample best DSR	1.000
OOS mean stability Sharpe	+0.72
OOS stability DSR	0.038
SR■ threshold (100 trials)	+2.51
Baseline-excess Sharpe (mean)	-1.126

Trials beating equal-weight	0%
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The in-sample best-val Sharpe of 6.74 is an order statistic across ~100 trials — the maximum of many noisy 5-episode validation estimates. The honest OOS Sharpe (mean across two retrained seeds on a 251-day held-out window) is 0.72, with DSR 0.04. The absolute Sharpe is predominantly beta exposure to a period when the equal-weight basket of the same 50 names earned Sharpe > 2.

6.3 Interpretation

The honest read of the rigor reports: **no strategy in the current project has credible, cost-adjusted, deflation-aware alpha** on this universe. The most promising factor (Momentum 12-1) has IC statistics consistent with the published literature, but its net-of-costs Sharpe collapses to +0.11 with a bootstrap 95% CI spanning zero. Equal-weight long-only beats every overlay. MARL absolute Sharpe is beta, not alpha.

This is a *research finding*, not a project failure. It identifies exactly what signal quality would be required to clear the bar, and provides the rigor scaffolding to test future candidates (residual momentum, short-interest changes, earnings-call sentiment, see §10).

7. Engineering highlights

- **JS/Python numerical parity to 10 decimal places.** Every JS-parity factor has a Python `compute_js()` that reproduces the frontend output bit-for-bit. Verified via `tests/fixtures/js_reference_output.json`.
- **Defensive numerics everywhere.** `safe_div`, `sanitize_number`, `validate_series`, `clamp` applied consistently across both Python backends and the JS frontend. No NaN or Infinity can propagate through the pipeline.
- **Parquet-store access pattern.** Per-ticker-per-year files with manifest validation, LRU-cached reads, strict range enforcement, and a quarantine subsystem for files that fail validator checks.
- **NSGA-II + Speciation.** Multi-objective selection (Sharpe, drawdown, turnover) with Jensen-Shannon-distance speciation prevents premature convergence.
- **FOMAML meta-learning.** Periodic first-order MAML updates on elite agents for faster regime adaptation.
- **Ticker-attention encoder.** Multi-head self-attention over 10 ticker slots in the actor-critic trunk to learn cross-ticker relationships.
- **Curriculum learning.** Four difficulty stages with progressively tightening tx costs, leverage limits, stop-loss thresholds, and episode length.
- **Walk-forward validator.** Anchored splits (train 2022–23, validate 2024, test 2025) with strict temporal isolation and overfitting-ratio reporting.
- **Deflation-aware reports.** Bailey & López de Prado DSR + stationary bootstrap + baseline-excess decomposition on both the factor and MARL sides.

- **618+ passing tests** across the three backends.

8. Live paper-trading execution

alphaforge-execution runs a daily trading loop against Alpaca paper-trading with live yfinance prices. The loop is: fetch prices → compute momentum composite ranking → risk-check (position size, exposure, turnover, circuit breakers for daily-loss and max-drawdown) → execute orders via broker ABC (PaperBroker with slippage for local sim; AlpacaBroker for live) → snapshot NAV/Sharpe/drawdown/win-rate into SQLite.

The executed strategy is an extraction of the momentum composite from the MARL environment's `_rank_tickers()`: 40% 5d momentum + 40% 21d momentum + 20% mean reversion, top-N equal-weight. Configuration lives in `configs/execution_config.yaml`. 106 tests cover broker interfaces, risk checks, circuit breakers, and the persistence layer.

9. Honest limitations

- **Survivorship bias.** Universe defined as of today, not point-in-time. Estimated 1–2%/yr upward bias on the equal-weight baseline.
- **No borrow costs.** Short-leg returns assume free unlimited borrow. Realistic borrow would add 20–100 bps/yr of drag on non-mega-cap short legs.
- **Static cost model.** Single commission + half-spread + impact parameters. Real impact scales with ADV participation; spread scales with volatility.
- **Small universe.** 50 tickers means quintile buckets are 10 names — cross-sectional IC t-stats are noisier than on a 500-name universe.
- **No risk model.** Returns not neutralised against sector or style factors. Reported alpha is partly explained by sector tilts.
- **MARL trial count under-reported.** The rigor script enumerates only generation-level trial rows; architecture search, curriculum choices, and reward-shape tuning are additional trials that should also deflate the DSR.
- **OOS window is 1 year for 2 seeds.** Insufficient statistical power to reject a zero-alpha null.

10. Roadmap — what moves this to a defensible result

The project's current posture is *rigorous infrastructure around textbook signals*. Converting that into *rigorous infrastructure around defensible signals* is the explicit next-phase work:

10.1 Strategy layer

- **Residual momentum** (Blitz-Huij-Martens 2011) — FF-5-residualised momentum, same universe, same pipeline. Expected to dominate plain 12-1 here.

- **One Tier-2 alt-data signal** — short-interest changes (FINRA, free) or earnings-call sentiment (FinBERT on transcripts). Differentiated for quant-researcher interviews.
- **Combination** — ERC / Bayesian IC-weighted blend of residual momentum + the alt-data signal. Target: combined DSR > max(individual DSRs).

10.2 Infrastructure

- **Point-in-time universe.** Move off of today's survivor set. Norgate or CRSP.
- **Sacred OOS lock.** 2023-01-01 → today held back from any tuning loop.
- **FF-5 residualisation layer.** Regress every strategy's returns on the Fama-French-5 factors + Carhart momentum to isolate true alpha.
- **Capacity analysis.** % of ADV consumed at \$10M / \$100M / \$1B AUM.
- **Borrow cost module.** Attach a per-ticker borrow curve to short legs.

Research philosophy. The value of the current project is not a winning strategy — it is a platform that makes winning and losing strategies *legibly* distinguishable. That property is rare in portfolio projects, and it is the platform on which the next phase of signal work will run.

Repository: *Quant Alpha* (AlphaForge). Generated by docs/build_alphaforge_pdf.py. Reproducible; re-run after any new factor or MARL rigor update to regenerate the PDF with the latest numbers pulled from the JSON artifacts.